Context-Aware Attentive Knowledge Tracing

Aritra Ghosh
University of Massachusetts Amherst
Amherst, MA
arighosh@cs.umass.edu

Neil Heffernan
Worcester Polytechnic Institute
Worcester, MA
nth@wpi.edu

Andrew S. Lan
University of Massachusetts Amherst
Amherst, MA
andrewlan@cs.umass.edu

ABSTRACT

Knowledge tracing (KT) refers to the problem of predicting future learner performance given their past performance in educational applications. Recent developments in KT using flexible deep neural network-based models excel at this task. However, these models often offer limited interpretability, thus making them insufficient for personalized learning, which requires using interpretable feedback and actionable recommendations to help learners achieve better learning outcomes. In this paper, we propose attentive knowledge tracing (AKT), which couples flexible attention-based neural network models with a series of novel, interpretable model components inspired by cognitive and psychometric models. AKT uses a novel monotonic attention mechanism that relates a learner's future responses to assessment questions to their past responses; attention weights are computed using exponential decay and a context-aware relative distance measure, in addition to the similarity between questions. Moreover, we use the Rasch model to regularize the concept and question embeddings; these embeddings are able to capture individual differences among questions on the same concept without using an excessive number of parameters. We conduct experiments on several real-world benchmark datasets and show that AKT outperforms existing KT methods (by up to 6% in AUC in some cases) on predicting future learner responses. We also conduct several case studies and show that AKT exhibits excellent interpretability and thus has potential for automated feedback and personalization in real-world educational settings.

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1 INTRODUCTION

Recent advances in data analytics and intelligent tutoring systems [32] have enabled the collection and analysis of large-scale learner data; these advances hint at the potential of *personalized* learning at large scale, by automatically providing personalized feedback [24] and learning activity recommendations [11] to each learner by analyzing data from their learning history.

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A key problem in learner data analysis is to predict future learner performance (their responses to assessment questions), given their past performance, which is referred to as the knowledge tracing (KT) problem [3]. Over the last 30 years, numerous methods for solving the KT problem were developed based on two common assumptions: i) a learner's past performance can be summarized by a set of variables representing their current latent *knowledge level* on a set of concepts/skills/knowledge components, and ii) a learner's future performance can be predicted using their current latent concept knowledge levels. Concretely, let t denote a set of discrete time indices, we have the following generic model for a learner's knowledge and performance

$$r_t \sim f(h_t), \quad h_t \sim g(h_{t-1}),$$

where $r_t \in \{0, 1\}$ denotes the learner's graded response to an assessment question at time step t, which is usually binary-valued (1 corresponds to a correct response and 0 corresponds to an incorrect one) and is observed. The latent variable h_t denotes the learner's current knowledge level and is not observed. $f(\cdot)$ and $g(\cdot)$ are functions that characterize how learner knowledge dictate their responses and how it evolves; they are sometimes referred to as the response model and the knowledge evolution model, respectively.

Earlier developments in KT methods before 2010 can be divided into two classes. The first class centered around the Bayesian knowledge tracing (BKT) method [19, 35] where knowledge (h_t) is a binary-valued scalar that characterizes whether or not a learner masters the (single) concept covered by a question. Since the response (r_t) is also binary-valued, the response and knowledge evolution models are simply noisy binary channels, parameterized by the guessing, slipping, learning, and forgetting probabilities. The second class centered around item response theory (IRT) models [16] and use these models (especially sigmoidal link functions) as the response model $f(\cdot)$; learner knowledge level is then modeled as real-valued vectors ($\mathbf{h}_t$) for questions that cover multiple concepts. Among these methods, the SPARFA-Trace method [13] used a simple affine transformation model as the explicit knowledge evolution model $g(\cdot)$. Other methods, e.g., additive factor models [1], performance factor analysis [22], the difficulty, ability, and student history (DASH) model [15], and a few recent methods including knowledge factorization machines [30] and an extension to the DASH model, the DAS3H model [2], used *hand-crafted features* such as the number of previous attempts, successes, and failures on each concept in their knowledge evolution model. Methods in both classes rely on expert labels to associate questions to concepts, resulting in excellent interpretability since they can effectively estimate the knowledge level of each learner on expert-defined concepts.

Recent developments in KT centered around using more sophisticated and flexible models to fully exploit the information contained

in large-scale learner response datasets. The deep knowledge tracing (DKT) method [23] was the first method to explore the use of (possibly deep) neural networks for KT by using long short-term memory (LSTM) networks [7] as the knowledge evolution model $g(\cdot)$. Since LSTM units are nonlinear, complex functions, they are more flexible than affine transformations and more capable of capturing nuances in real data.

The dynamic key-value memory networks (DKVMN) method extended DKT by using an external memory matrix ($\mathbf{H}_t$) to characterize learner knowledge [36]. The matrix is separated into two parts: a static, "key" matrix that contains the fixed representation of each concept, and a dynamic, "value" matrix that contains the evolving knowledge levels of each learner on each concept. DKVMN also uses separate "read" and "write" processes on this external matrix for the response and knowledge evolution models; these processes make it even more flexible than DKT. DKT and KVMN reported state-of-the-art performance in predicting future learner performance [9] and have been the benchmark for new KT methods.

The self-attentive knowledge tracing (SAKT) method [18] is the first method to use attention mechanisms in the context of KT. Attention mechanisms are more flexible than recurrent and memory-based neural networks and have demonstrated superior performance in natural language processing tasks. The basic setup of SAKT has many similarities to the Transformer model [29], an effective model for many sequence-to-sequence prediction tasks. However, we observe that SAKT does not outperform DKT and DKVMN in our experiments; see Section 4 for details. Possible reasons for this include i) unlike in language tasks where strong long-distance dependencies between words are more common, the dependence of future learner performance on the past is likely restricted to a much shorter window, and ii) the sizes of learner response datasets are several magnitudes lower than natural language datasets and are less likely to benefit from highly flexible and large-scale attention models.

More importantly, no existing KT method truly excels at both future performance prediction and interpretability. Early KT methods exhibit excellent interpretability but do not provide state-of-the-art performance on future learner performance prediction. Recent KT methods based on deep learning excel at that but offer limited interpretability. Therefore, these KT methods do not fully meet the needs of personalized learning, which requires not only accurate performance prediction but also the ability to provide automated, interpretable feedback and actionable recommendations to help learners achieve better learning outcomes.

1.1 Contributions

For the task of predicting the learner's response to the current question, we propose the attentive knowledge tracing (AKT) method, which uses a series of attention networks to draw connections between this question and every question the learner has responded to in the past. We summarize our key innovations below:

(1) Contrary to existing attention methods that use raw question and response embeddings, we put raw embeddings into context and use *context-aware representations* of past questions and responses by taking a learner's entire practice history into account.
(2) Inspired by cognitive science findings on the mechanisms of forgetting, we propose a novel *monotonic attention mechanism* that uses an exponential decay curve to down weight the importance of questions in the distant past. We also develop a context-aware measure to characterize the time distance between questions a learner has responded to in the past.
(3) Leveraging the Rasch model, a simple and interpretable IRT model, we use a series of *Rasch model-based embeddings* to capture individual differences among questions without introducing an excessive amount of model parameters.

We conduct a series of experiments on several benchmark real-world educational datasets comparing AKT to state-of-the-art KT methods. Our results show that AKT (sometimes significantly) outperforms other KT methods in predicting future learner performance. Further, we perform ablation studies on each of the key AKT model components to demonstrate their value. We also perform several case studies to show that AKT exhibits excellent interpretability and has the potential for automated feedback and practice question recommendation, both key requirements of personalized learning[1].

2 KNOWLEDGE TRACING PROBLEM SETUP

Each learner's performance record consists of a sequence of questions and responses at each discrete time step. For learner i at time step t, we denote the combination of the question that they answered, the concept this question covers, and their graded response as a tuple, (q_t^i, c_t^i, r_t^i), where $q_t^i \in \mathsf{N}^+$ is the question index, $c_t^i \in \mathsf{N}^+$ is the concept index, and $r_t^i \in \{0, 1\}$ is the response. Under this notation, $(q_t^i, c_t^i, 1)$ means learner i responded to question q_t^i on concept c_t^i correctly at time t. We note that this setup is different from some prior works on deep knowledge tracing that often ignore the question index and summarize learner performance as (c_t^i, r_t^i). This choice was made to avoid overparameterization; see Section 3.3 for a detailed analysis. In the following discussions, we omit the superscript i as we discuss how to predict future performance for a single learner. Given their past history up to time $t-1$ as $\{(q_1, c_1, r_1), \ldots, (q_{t-1}, c_{t-1}, r_{t-1})\}$, our goal is to predict their response r_t to question q_t on concept c_t at the current time step, t.

2.1 Question and Response Embeddings

Following previous work [36], we use real-valued embedding vectors $\mathbf{x}_t \in \mathsf{R}^D$ and $\mathbf{y}_t \in \mathsf{R}^D$ to represent each question and each question-response pair (q_t, r_t), respectively. $\mathbf{x}_t$ characterizes information about questions, and $\mathbf{y}_t$ characterizes the knowledge learners acquire by responding to questions, with two separate embeddings for correct and incorrect responses, respectively. D denotes the dimension of these embeddings. Therefore, let Q denote the number of questions, there are a total of Q question embedding vectors and $2Q$ question-response embedding vectors. In most real-world educational settings, the question bank is considerably larger than the set of concepts and many questions are assigned to very few learners. Therefore, the majority of existing KT methods use concepts to index questions to avoid overparameterization; all questions covering the same concept are treated as a single question. In this case, $q_t = c_t$ and $Q = C$.

[1] Source code and datasets will be available at https://github.com/arghosh/AKT

3 THE AKT METHOD

The AKT method consists of four components: two self-attentive encoders, one for questions and one for knowledge acquisition, a single attention-based knowledge retriever, and a feed-forward response prediction model; Figure 1 visualizes the AKT method and its connected components.

We use the two self-attentive encoders to learn context-aware representations of the questions and responses. We refer to the first encoder as the *question encoder*, which produces modified, contextualized representations of each question, given the sequence of questions the learner has previously practiced on. Similarly, we refer to the second encoder as the *knowledge encoder*, which produces modified, contextualized representations of the knowledge the learner acquired while responding to past questions. Alternatively, we could use raw embeddings of questions and responses similar to prior work. We found that the context-aware representation performs better in most datasets. We refer to the knowledge evolution model as the *knowledge retriever*, which retrieves knowledge acquired in the past that is relevant to the current question using an attention mechanism. Finally, the response prediction model predicts the learner's response to the current question using the retrieved knowledge. The AKT method is motivated by three intuitions rooted in cognitive science and psychometrics; we will detail these intuitions in what follows.

3.1 Context-aware Representations and The Knowledge Retriever

As introduced above, we use two encoders in our model. The question encoder takes raw question embeddings $\{\mathbf{x}_1, \ldots, \mathbf{x}_t\}$ as input and outputs a sequence of context-aware question embeddings $\{\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_t\}$ using a monotonic attention mechanism (detailed in the next subsection). The context-aware embedding of each question depends on both itself and the past questions, i.e., $\hat{\mathbf{x}}_t = f_{\text{enc}_1}(\mathbf{x}_1, \ldots, \mathbf{x}_t)$. Similarly, the knowledge encoder takes raw question-response embeddings $\{\mathbf{y}_1, \ldots, \mathbf{y}_{t-1}\}$ as input and outputs a sequence of actual knowledge acquired $\{\hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_{t-1}\}$ using the same monotonic attention mechanism. The context-aware embedding of acquired knowledge depend on the learner's response to both the current question and past questions, i.e., $\hat{\mathbf{y}}_{t-1} = f_{\text{enc}_2}(\mathbf{y}_1, \ldots, \mathbf{y}_{t-1})$.

The choice of using context-aware embeddings rather than raw embeddings reflects our first intuition: *the way a learner comprehends and learns while responding to a question depends on the learner.* These modified representations reflect each learner's actual comprehension of the question and the knowledge they actually acquire, given their personal response history. This model choice is motivated by the intuition that for two learners with different past response sequences, the way they understand the same question and the knowledge they acquire from practicing on it can differ.

The knowledge retriever takes the context-aware question and question-response pair embeddings $\hat{\mathbf{x}}_{1:t}$ and $\hat{\mathbf{y}}_{1:t-1}$ as input and outputs a retrieved knowledge state $\mathbf{h}_t$ for the current question. We note that in AKT, the learner's current knowledge state is also context-aware since it depends on the current question they are responding to; this model choice is different from that in most existing methods, including DKT. We also note that the knowledge retriever can only use information on the past questions, the learner's responses to them, and the representation of the current question, but not the learner's response to the current question, i.e., $\mathbf{h}_t = f_{\text{kr}}(\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_t, \hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_{t-1})$. The response prediction model uses the retrieved knowledge to predict the current response.

3.2 The Monotonic Attention Mechanism

We use a modified, monotonic version of the scaled dot-product attention mechanism for the encoders and the knowledge retriever. We start by briefly summarizing the original scaled dot-product attention mechanism. Under this framework, each encoder and the knowledge retriever has a key, query, and value embedding layer that maps the input into output queries, keys, and values of dimension $D_q = D_k$, D_k, and D_v, respectively. Let $\mathbf{q}_t \in \mathbb{R}^{D_k \times 1}$ denote the query corresponding to the question the learner responds to at time t, the scaled dot-product attention values $\alpha_{t,\tau}$ are computed using the softmax function [5] as

$$\alpha_{t,\tau} = \text{Softmax}(\frac{\mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{D_k}}) = \frac{\exp(\frac{\mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{D_k}})}{\sum_{\tau'} \exp(\frac{\mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{D_k}})} \in [0, 1].$$

The output of the scaled dot-product attention mechanism is then given by $\sum_\tau \alpha_{t,\tau} \mathbf{v}_\tau \in \mathbb{R}^{D_v \times 1}$. $\mathbf{k}_\tau \in \mathbb{R}^{D_k \times 1}$ and $\mathbf{v}_\tau \in \mathbb{R}^{D_v \times 1}$ denote the key and value for the question at time step τ, respectively. Depending on the specific component, the output depends either on both the past and the current ($\tau \leq t$ for the question and knowledge encoders) or only the past ($\tau < t$ for the knowledge retriever).

Both encoders employ the self-attention mechanism, i.e., $\mathbf{q}_t$, $\mathbf{k}_t$, and $\mathbf{v}_t$ are computed using the same input; the question encoder uses $\{\mathbf{x}_1, \ldots, \mathbf{x}_t\}$ while the knowledge encoder uses $\{\mathbf{y}_1, \ldots, \mathbf{y}_{t-1}\}$. The knowledge retriever, on the other hand, does not use self-attention. As shown in Fig. 1, at time step t, it uses $\hat{\mathbf{x}}_t$ (the modified embedding of the current question), $\{\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_{t-1}\}$ (the context-aware embeddings of past questions), and $\{\hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_{t-1}\}$ (the context-aware embeddings of past question-response pairs) as input to generate the query, keys, and values, respectively. We note that SAKT uses question embeddings for mapping queries whereas response embeddings for key and value mapping. In our experiments, we have found that using question embeddings for mapping both queries and keys is much more effective.

However, this basic scaled dot-product attention mechanism is not likely going to be sufficient for KT. The reason is that learning is temporal and memories decay [21]; a learner's performance in the distant past is not as informative as recent performance when we are predicting their response to the current question. Therefore, we develop a new monotonic attention mechanism that reflects our second intuition: *when a learner faces a new question, past experiences i) on unrelated concepts and ii) that are from too long ago are not likely to be highly relevant.* Specifically, we add a multiplicative exponential decay term to the attention scores as:

$$\alpha_{t,\tau} = \frac{\exp(s_{t,\tau})}{\sum_{\tau'} \exp(s_{t,\tau'})},$$